

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025/2026

OCTOBER EXAMINATION

AMIS1012 ETHICS IN COMPUTING

WEDNESDAY, 8 OCTOBER 2025

TIME: 9.00 AM – 11.00 AM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

DIPLOMA IN INFORMATION TECHNOLOGY

DIPLOMA IN INFORMATION SYSTEMS

DIPLOMA IN SOFTWARE ENGINEERING

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AMIS1012 ETHICS IN COMPUTING**Question 1**

- a) Provide a scenario related to information technology to explain the differences among morals, ethics, and law. (9 marks)
- b) Discuss each of the following active attacks and provide an example for each:
 - (i) Virus (2 marks)
 - (ii) Rootkit (2 marks)
 - (iii) Distributed Denial of Service (DDoS) attack (2 marks)
- c) Explain **FIVE (5)** common types of spam encountered in digital communication. For each type, provide a clear description of its characteristics and how it typically functions. (10 marks)

[Total: 25 marks]

Question 2

PQR Sdn. Bhd. is a retail business that sells computer and hardware products, including desktops, laptops, keyboards, mice, and other related items. The company recently upgraded its point-of-sale (POS) system to a cloud-based solution. During the upgrade process, the business owner instructed the IT team to conduct a security risk assessment, with an emphasis on ensuring the security of customer payment data.

- a) Explain how the IT team at PQR Sdn. Bhd. would apply the process of a security risk assessment to achieve this goal. (3 marks)
- b) Outline the **EIGHT (8)** steps in a general security risk evaluation process that should be conducted by the IT team from PQR Sdn. Bhd. (8 marks)
- c) Based on the case above, identify the main asset that appears important to PQR Sdn. Bhd. (2 marks)
- d) CIA triad principles are essential to be adopted in the new upgraded system. Discuss the **THREE (3)** CIA triad principles and provide a scenario for each principle to demonstrate how it should be applied in PQR Sdn. Bhd. (12 marks)

[Total: 25 marks]

Question 3

“Identifying and eliminating a defect early in the software development process will cost less than removing a defect in software that has already been shipped to consumers.”

- a) Do you agree with the statement above? Provide relevant supporting justifications to elaborate on your point of view. (5 marks)

AMIS1012 ETHICS IN COMPUTING**Question 3 (Continued)**

- b) A software development team decided to use dynamic testing on the system they are developing. Explain the following types of dynamic testing and provide a scenario illustrating how the software development team can carry out each testing.
- (i) Black-box testing (4 marks)
 - (ii) White-box testing (4 marks)
- c) You are a software designer tasked with planning the development of a new mobile application for secure online payment transactions. Based on this scenario, explain how you would apply **FOUR (4)** characteristics of a high-quality software product in your design considerations to ensure the app meets user needs. (12 marks)

[Total: 25 marks]

Question 4

Edward is a freelance program developer. He was hired by XYZ company to help develop an animated cartoon character to assist user in using their application. When designing the cartoon character, Edward created a dwarf cartoon character that closely resembles one of the Seven Dwarfs from Snow White, originally created by the Walt Disney Company.

- a) Based on the scenario above, explain how the Walt Disney Company would apply their Intellectual Property Rights (IPR) to protect their contributions throughout the production and distribution process. Provide a specific example for each. (5 marks)
- b) Based on the scenario above, if the Walt Disney Company decided to take legal action against Edward, describe how the company could specifically use **TWO (2)** relevant types of IPR to address Edward's infringement. For each party, provide **ONE (1)** concrete action or claim they could initiate. (10 marks)
- c) Briefly explain **FIVE (5)** issues related to intellectual property protection. (10 marks)

[Total: 25 marks]